GlucoTwin

A Reliability-Aware Digital Twin for 120-Minute Glucose Forecasting & Adverse Event Mitigation

Competition: Happiest Health — Digital Twin Challenge 2026

Summit: Reimagining & Reforming Healthcare in India Summit 2026, Bengaluru

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Repository: https://github.com/Raju-09/GlucoTwin • **License:** MIT Open Source

1. The Healthcare Challenge in India

101 Million Patients: The Critical Need for Proactive Glycemic Management

- **Global Diabetes Epicenter:** India has over 101 million diagnosed individuals with diabetes (IDF Diabetes Atlas, 2023), with millions undiagnosed in tier-2/3 cities and rural hubs.
- **The Fatal Flaw of Modern CGMs:** Continuous Glucose Monitors generate readings every 5 minutes, but current alert algorithms are entirely *reactive*. They fire alarms only after blood glucose breaches dangerous thresholds (<70 mg/dL for hypoglycemia, >180 mg/dL for hyperglycemia), when metabolic stress has already commenced.
- **The Proactive 2-Hour Window:** A validated 120-minute forward forecast provides actionable lead time for dietary adjustment, physical activity, or preemptive clinical triage before acute excursions occur.
- **Data Privacy Sandbox Compliance:** Built and evaluated using in-silico synthetic cohorts and open-source formats in strict compliance with India's DPDP Act and global HIPAA frameworks.

'Reactive alarms notify patients of emergencies; proactive digital twins prevent them.'

2. Mathematical Formulation: Patient Digital State

Decoupling Static Phenotype from Dynamic Causal History

GlucoTwin formally represents the virtual patient at any discrete time step *t* as a triad:

$$S(t) = (\theta_p , h_t , \delta_t)$$

Component	Physiological & Algorithmic Meaning	Features Included
θ_p **Static Phenotype**	Longitudinal EHR context, clinical intake, demographic & baseline metabolic indicators.	Age, Sex, BMI, Baseline HbA1c, Diabetes Duration, Historical Fasting Glucose, Metformin/SGLT2i/Dawn Flags (9 feats)
h_t **Dynamic History**	Causal sliding window of sensor observations (120 min lookback = 24 steps at 5-min cadence).	CGM Lags 0..23, Rolling Mean/Std (1h, 2h), Discrete Velocity dG/dt, Acceleration, Circadian Sin/Cos, Wearable Telemetry
δ_t **Twin Integrity**	Real-time stream quality, latency, telemetry jitter, and fail-closed forecast gating state.	Integrity State (AVAILABLE, DEGRADED, WITHHELD), Reason Codes, Telemetry Freshness, Quality Score %

3. End-to-End System Architecture

Three-Tier Architecture: Ingestion -> Verification -> Serving

- **Tier 1 — Preprocessing & Sensor Physics:** Continuous CGM readings pass through an interstitial physiological delay model (τ=10 min), AR(1) autocorrelated sensor noise, and rate clamping ($|dG/dt| \leq 3.8$ mg/dL/min) to reflect true subcutaneous biology.
- **Tier 2 — Causal Multimodal Feature Pipeline:** Combines 24 CGM lags, rolling kinematics, circadian encodings, and EHR intake into a 56-feature vector. Supervised training windows use strict chronological boundaries with a 120-minute gap buffer to guarantee zero future data leaks.
- **Tier 3 — HistGradientBoosting Core & Reliability Gating:** High-efficiency tree regressor trained on 224,681 windows, coupled to a finite-state machine that evaluates input admissibility before returning a prediction.
- **Tier 4 — REST API & Clinician Dashboard:** Sub-10ms FastAPI backend powering a Streamlit UI with 14-day timeline scrubbing, state inspector, and one-click failure injection.

Full vector diagram available at: docs/architecture_diagram.pdf

4. Core Differentiator: Fail-Closed Reliability Gating

The Digital Twin Knows When NOT to Trust Its Predictions

In clinical machine learning, an overconfident wrong prediction during hardware failure is catastrophic. GlucoTwin enforces fail-closed behavior across three formal states:

Status Level	Trigger Conditions	System Behavior
■ **AVAILABLE**	Nominal stream, freshness < 15 min, complete 24-step history, values in [40, 400] mg/dL.	120-min point forecast displayed with high-confidence green badge.
■ **DEGRADED**	Mild telemetry latency (15–20 min) or minor packet loss; historical context intact.	Forecast displayed alongside advisory warning flag & degraded reason code.
■ **WITHHELD**	Stale stream (>20 min), severe gap (>30 min), sensor dropout, or jump > 4.0 mg/dL/min.	**Output suppressed. No number returned.** Explicit error code issued (e.g. ERR_SENSOR_DROPOUT).

Utility Analysis: Our risk-coverage audit proves that withholding 12% of corrupted inputs suppresses severe prediction errors (>30 mg/dL) and protects clinical fidelity.

5. Multimodal Ablation Study (N=60 Patients)

Evaluated on 43,888 Held-Out Test Windows Under Strict Chronological Hold-Out

Configuration	Input Streams & Features	Test MAE	Δ MAE vs Ref	95% Bootstrap CI
Persistence Baseline	Naive y(t+120) = y(t)	53.25 mg/dL	+34.94 mg/dL	—
Config A (CGM Only)	24 Lags + Rolling Kinematics (41 feats)	18.31 mg/dL	Reference	—
Config B (CGM + Static EHR)	Demographics + Labs + Fasting Lab (50 feats)	**17.61 mg/dL**	**–0.70 mg/dL**	**[+0.49, +0.96]** (p < 10■■)
Config C (CGM + Wearables)	Decoupled HR, HRV, Steps, Sleep (47 feats)	18.32 mg/dL	+0.01 mg/dL	[–0.08, +0.07] (p = 0.95)
Config D (Full Two-Stream)	EHR + CGM + Wearables (56 feats)	**17.67 mg/dL**	**–0.64 mg/dL**	[+0.44, +0.88]

• **EHR Gain:** Static EHR context provides a statistically validated error reduction (–0.70 mg/dL) across same-patient trajectories.
• **Wearable Negative Finding:** Activity telemetry provided negligible incremental gain under this simulator; reported honestly rather than fabricating artificial gains.

6. Breaking Our Own Claims: Leave-One-Patient-Out (LOPO)

Cross-Validation Across 60 Patients (224,681 Windows)

To test whether the model learned generalizable human physiology or merely memorized synthetic patient identity, we executed a complete Leave-One-Patient-Out (LOPO) cross-validation and a permuted-EHR control:

Evaluation Mode	Mean LOPO MAE	Delta vs CGM-Only	Scientific Verdict
CGM Only	17.65 mg/dL	Reference	Solid kinematic time-series tracking.
CGM + Real EHR	**17.65 mg/dL**	+0.01 mg/dL (p = 0.956)	**No transfer benefit on unseen patients.** EHR acts as same-patient anchor.
CGM + Permuted EHR	**20.47 mg/dL**	**+2.82 mg/dL degradation**	**Catastrophic failure.** Confirms model relies on true patient identity.

'A system that knows the boundary of its own evidence is far more trustworthy in healthcare than one claiming 99% universal accuracy.'

7. Clinical Adverse Event Classification (t+120 min)

Catching Postprandial Spikes 2 Hours Ahead & Suppressing Alarm Fatigue

Adverse Event	Model	Events (N)	Sensitivity	Specificity	Precision	F1-Score	False Alarms/Day
Hyperglycemia (>180 mg/dL)	Persistence Baseline	12,246	26.6%	70.4%	25.8%	0.262	61.4
Hyperglycemia (>180 mg/dL)	**GlucoTwin (Two-Stream)**	12,246	**76.2%**	**90.6%**	**75.8%**	**0.760**	**19.6 (–68%)**
Hypoglycemia (<70 mg/dL)	Persistence Baseline	71	12.7%	99.9%	12.7%	0.127	0.4
Hypoglycemia (<70 mg/dL)	**GlucoTwin (Two-Stream)**	71	0.0%	100.0%	0.0%	0.000	0.0

• **Postprandial Excursions:** Catches nearly 3× more hyperglycemic events 2 hours in advance while reducing false alarms from 61.4 to 19.6/day (a **68% reduction**).
• **Hypoglycemia Tail Limitation:** In adult T2D, hypoglycemia is a rare tail event (<0.2% prevalence). Standard MSE regression regresses towards the mean (~150 mg/dL) and misses rare nadirs below 70 mg/dL; documented as a future milestone for asymmetric loss weighting.

8. Clinician Dashboard & Engineering Maturity

Production-Quality Engineering Ready for Triage

- **Doctor's Interactive Dashboard:** Streamlit-based UI featuring 14-day continuous time scrubbing across 60 patients, clear data provenance badges ([Observed EHR], [Sensor], [Kinematic], [Model]), and unclamped discrete velocity tracking.
- **Live Telemetry Stress Testing:** Real-time sidebar allows clinicians and evaluators to inject sensor dropouts, 20-minute latency delays, or non-physiological spikes (400 mg/dL) to verify fail-closed gating behavior live.
- **Sub-10ms Serving Latency:** Measured live over 100 consecutive API inferences on commodity hardware: **Median: 7.79 ms, Mean: 8.11 ms, p95: 11.14 ms, p99: 13.26 ms**.
- **Containerized Deployment:** Fully reproducible via docker-compose up in under 60 seconds with 69 unit tests passing with 100% test coverage.

Live demo available locally via: streamlit run frontend/streamlit_app.py

9. Responsible AI, Limitations & Clinical Translation

Transparent Boundaries and the Path to Indian Healthcare Impact

- **Explicit Non-Goals:** Research prototype; NOT an FDA/CDSCO-cleared medical device. Evaluated on in-silico synthetic cohorts and must not be used to direct autonomous insulin dosing or clinical therapy.
- **Phase 1 (Completed):** In-silico digital twin architecture, multi-stream feature engineering, fail-closed reliability FSM, and LOPO cross-validation.
- **Phase 2 (Next Milestone):** Prospective transfer validation on clinical research datasets (DiaTrend, OhioT1DM) under data transfer agreements.
- **Phase 3 (Long-Term Vision):** Low-cost edge deployment in Indian primary healthcare centers (PHCs) to assist community physicians in proactive lifestyle counseling.

GlucoTwin: Forecast Earlier. Intervene Smarter. Fail Closed When Uncertain.

Thank you. Team Synapse — Sammeta Siva Prasad Raju & Bedadakota Devi Chandrika